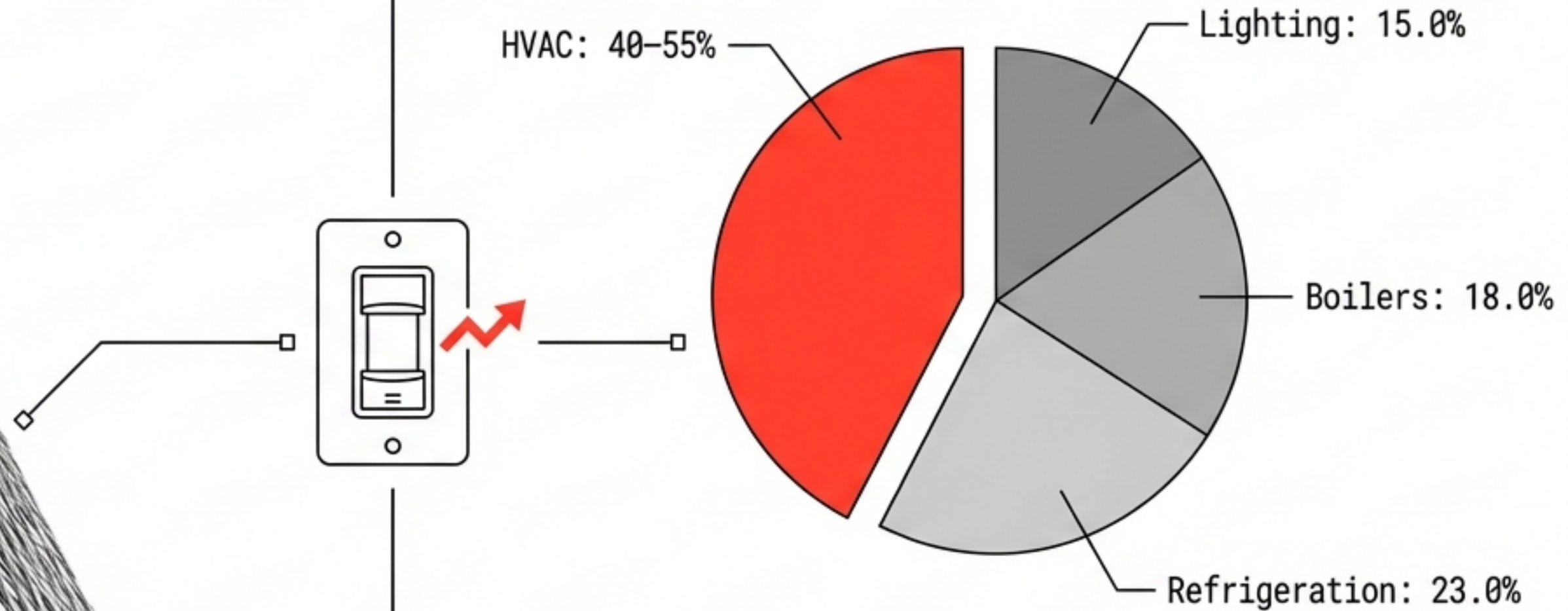


THE BLIND SPOT IN BUILDING AUTOMATION

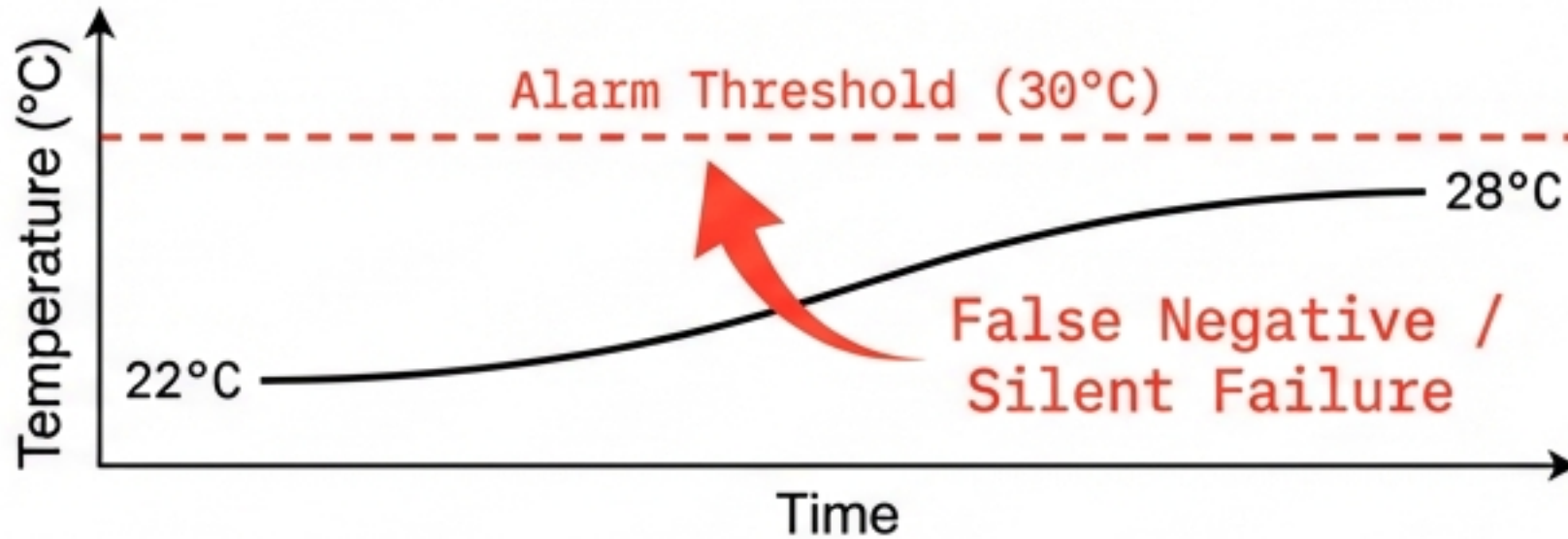


THE SYMPTOM:

1. HVAC dominates consumption (40-55% of total energy).
2. The Multiplier Effect: 1°C sensor deviation = 10-15% energy spike.
3. Market Scale: India Smart Building market growing at 24.2% CAGR.

Failure isn't a Crash. It's a Drift.

DIAGRAM A: The Old Way

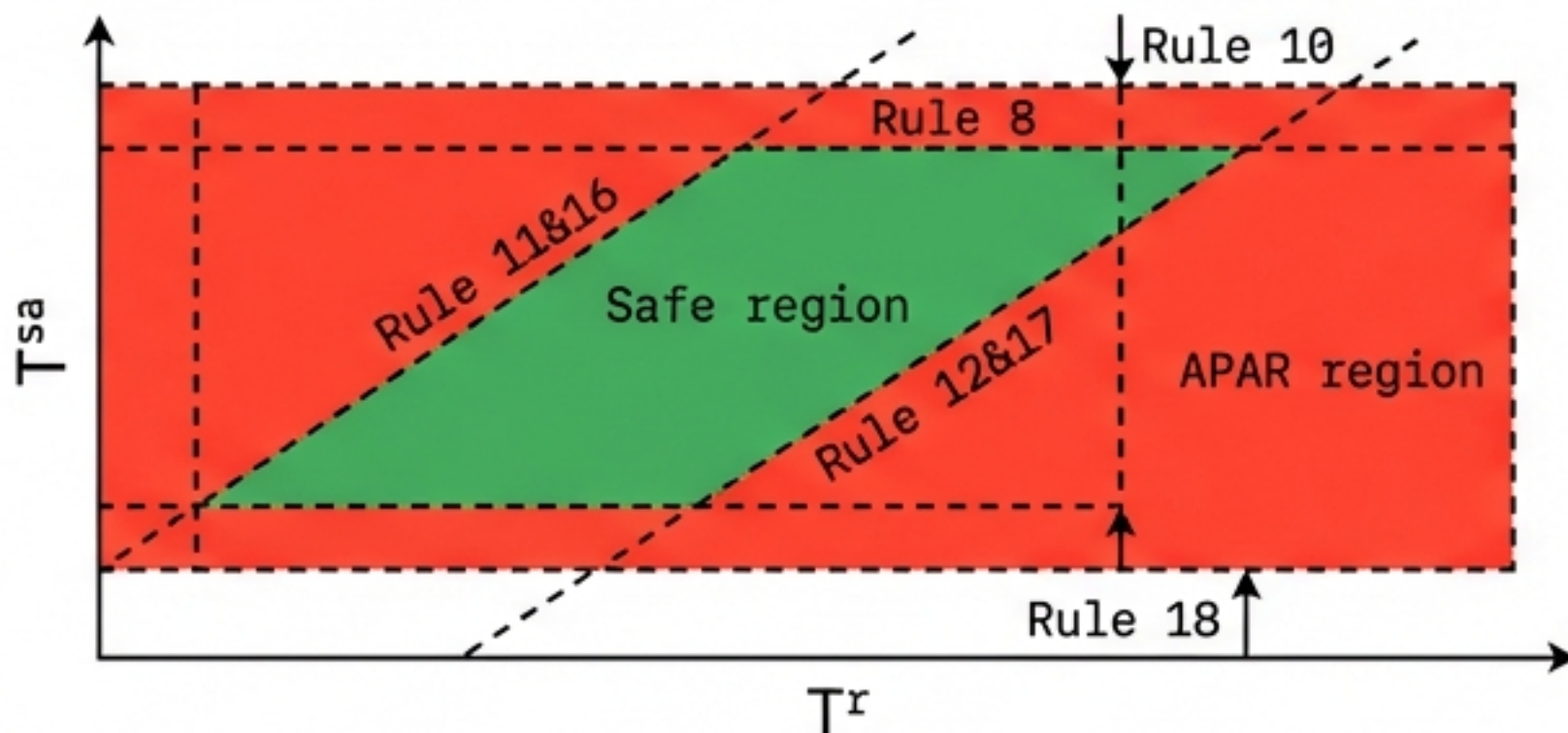


THE DIAGNOSIS:

Current defenses rely on static thresholds (If $T > 30^{\circ}\text{C}$, Alert).

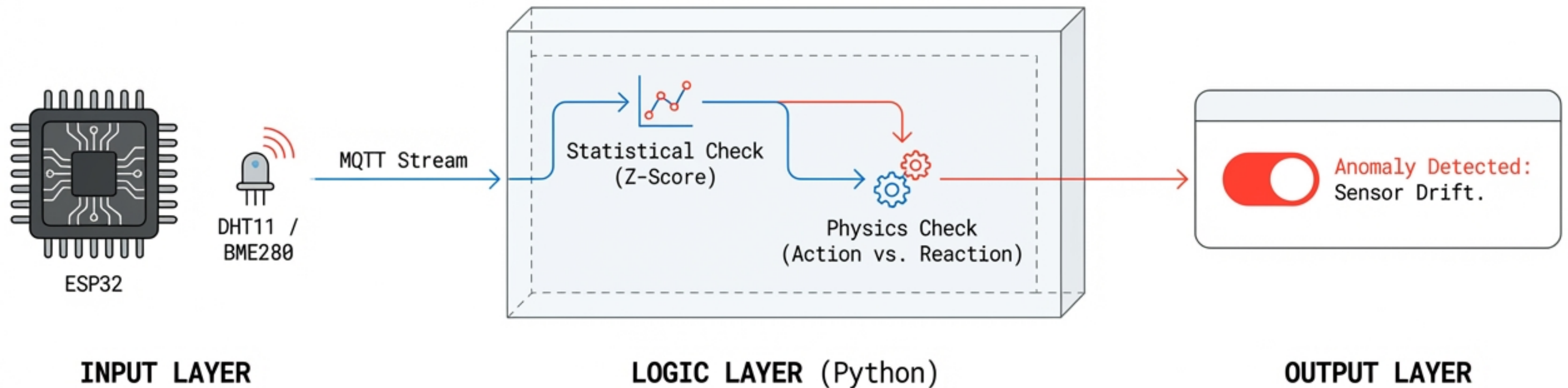
Attacks and sensor drift occur INSIDE the plausible range (e.g., $22^{\circ}\text{C} \rightarrow 25^{\circ}\text{C}$).

DIAGRAM B: The Reality



The AI trusts this false data and optimizes for a phantom environment, maximizing cooling for heat that doesn't exist.

Trust, But Verify: The Consistency Monitoring Layer (CML)



THE PRESCRIPTION: We do not replace the automation. We supervise it. Using low-cost edge nodes (ESP32) and Python-based logic to verify if sensor data obeys the laws of physics.

Grounded in Research, Critical for the Future

ACADEMIC VALIDATION:

1. **False Data Injection Attacks against Building Automation Systems.**
(Source: Cash et al., arXiv)
'Manipulated readings increase energy costs while remaining undetected.'
2. **Adversarial Resilience in AI Models.**
(Source: MarketsandMarkets / NIST)
'AI maintains high confidence even on spoofed input data.'

Real-World Applications



Commercial IT Parks
(High HVAC Load)



Smart Hospitals
(Critical Environments)



Cold Chain Logistics
(Pharma/Food Safety)

“Smart buildings rarely fail loudly. They fail quietly—and that is where the real risk lies.”